

StegoShield

A smart web platform for encrypted steganography and AI-based steganalysis

Project Proposal

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1. EXECUTIVE SUMMARY

StegoShield is a Python-based web application designed for covert, end-to-end encrypted communication and defensive steganalysis. Traditional encryption protects message confidentiality but leaves metadata exposed, signaling to attackers or surveillance systems that a secret exchange is taking place. StegoShield addresses this by combining AES-GCM symmetric encryption, X25519 public-key exchange, LSB steganography, a novel Pre-Flight Cover Image Risk & Capacity Checker, and a Machine Learning Steganalysis Detector into a single dashboard. The system empowers users to analyze cover images for detectability risks before embedding data while providing defenders with automated tools to spot pixel-level anomalies.

2. PROBLEM STATEMENT & MOTIVATION

Standard communication privacy relies heavily on cryptography. However, in high-surveillance environments, the mere presence of encrypted text can draw suspicion. Steganography mitigates this by concealing the secret message within an innocent carrier file, such as a PNG image.

However, current steganographic tools suffer from three major vulnerabilities:

1. **Poor Image Selection:** Users often choose smooth, low-variance images (e.g., clear skies) that make micro-pixel changes statistically obvious, leading to easy detection by steganalysis tools.
2. **Uncertain Payload Capacities:** Users do not know how much data can be safely hidden before visual or statistical degradation occurs.
3. **Lack of Defensive Assessment:** Most suites focus exclusively on hiding or detecting, rarely integrating pre-embedding security auditing alongside AI-driven detection.

StegoShield bridges these gaps by combining proactive risk scoring, secure steganographic embedding, and defensive AI detection.

3. KEY OBJECTIVES & SCOPE

- **Robust Cryptography:** Secure payloads with AES-256-GCM using pre-shared passwords (PBKDF2) or X25519 Elliptic-Curve Diffie-Hellman (ECDH) key exchange.
- **Pre-Flight ML Risk Audit:** Analyze candidate images before encoding to compute maximum safe payload capacity and predict an AI detectability risk score using machine learning regression.
- **Keyed Pseudorandom Embedding:** Scatter encrypted payload bits across image channels using key-derived pseudorandom permutations to defeat sequential LSB extraction.
- **AI-Driven Steganalysis:** Train a machine learning classifier (Random Forest / Light CNN) using LSB bit-plane statistics and spatial co-occurrence matrices to flag stego-images.
- **Streamlit Dashboard:** Deliver a clean 4-module web platform (Keys, Pre-Flight Audit, Hide & Extract, AI Detector).

4. TECHNICAL METHODOLOGY

Module A: Encryption & Key Management

Messages are authenticated and encrypted using AES-256-GCM. Keying is handled via two modes:

- *Pre-Shared Password Mode:* Uses PBKDF2 with a salt to derive symmetric keys.
- *Asymmetric Mode:* Uses X25519 public keys with HKDF key derivation, allowing secure communication without pre-shared passwords.

Module B: Pre-Flight Cover Image Checker & Risk Scoring Engine

Before embedding, the application evaluates the uploaded carrier image using a dual approach:

- **Max Capacity Calculation:** Computes exact safe payload limits in bytes based on image dimensions and bit-plane allocations.
- **ML-Based Detectability Risk Score:** A trained machine learning regression model (Random Forest Regressor) that extracts image statistical features to predict the exact probability score that an AI detector will successfully flag the embedded message. This gives users an instant pre-flight "Risk Rating (Low/Medium/High)" before they commit to hiding data.

Module C: Steganographic Embedding & Extraction

The encrypted payload (header + salt/nonce + ciphertext + authentication tag) is scattered across pixel channels using a key-seeded pseudorandom permutation generator. Un-keyed extraction attempts fail safely, and tampered images trigger AES-GCM tag verification failures.

Module D: Defensive AI Steganalysis Detector

A machine learning pipeline trained on clean and stego-image pairs generated at various payload ratios (5%, 10%, 25%, 50%). Features such as LSB-plane variance, chi-square attack statistics, and adjacent pixel differences are extracted to output a probability score indicating whether hidden data is present.

5. EXPERIMENTAL & EVALUATION PLAN

- **Imperceptibility & Visual Quality:** Evaluate Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) across 5%, 10%, 25%, and 50% payload ratios to ensure visual fidelity.
- **Detector Performance:** Benchmark ML classification Accuracy, Precision, Recall, and ROC curves against sequential versus pseudorandom LSB embedding.
- **Channel Robustness:** Test stego-image survivability across raw file/document transfers versus lossy social media recompression to document real-world transport constraints.
- **Security & Integrity Testing:** Validate authentication tag failures under bit-tampering and verify safe failure modes during incorrect key entries.